THE LAWS OF SOFTWARE TEAMS

Chesterton's fence

Do not remove a fence until you know why it was put up

swipe →

Ivan Rykov

01 THE HOOK

It looks useless
So you delete it

config/payments.yaml

```
timeout_ms: 5000
retries: 3
batch_delay_ms: 37
region: eu-west
```

no comment

delete line

the quick fix

illustration

02 THE FENCE

Chesterton's fence, across a road

G. K. Chesterton · The Thing · 1929



the road

THE QUICK REFORMER

"I don't see the use of this;
let us clear it away"

quoted from "The Drift from Domesticity"

Go away and think

THE WISER REFORMER

"If you don't see the use of it,
I certainly won't let you clear it away.
Go away and think. Then, when you can
come back and tell me that you do see
the use of it, I may allow you
to destroy it"

G. K. Chesterton, The Thing (1929), "The Drift from Domesticity"

Find out why first
Then decide

In software, fences are everywhere

```
time.sleep(2)
```

why: a race with the bank's API

```
if customer_id == 4417:
```

why: a contract with one large client

Manual approval step

why: an old audit finding

Each one had a reason

illustration

Knight Capital reused an old flag

AUGUST 1, 2012 · NEW CODE

It reused a flag that once switched on an old function, “Power Peg”

8 SERVERS · one missed the update



On that server, the flag woke up old code, unused since 2003

$460M+

lost in about 45 minutes

over 4 million executions in 154 stocks

SEC order on Knight Capital Americas, October 2013

Not: keep every fence
But: find out first



Ask before deleting
Explain when building

BEFORE YOU DELETE

- check the history
- ask the owner
- look for the test that breaks

WHEN YOU BUILD ONE · A DECISION RECORD

`ADR-014 · batch_delay_ms: 37`

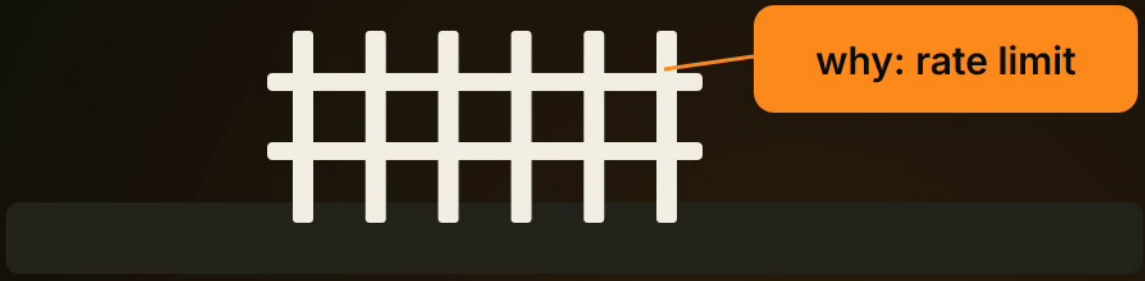
Why: the bank API limits bursts

Remove when: the bank ships v2

Owner: payments team

illustration · ADR = architecture decision record

Every fence has a story



Find it before you remove it

TAKEAWAY

Every fence has a story
Find it before you remove it

Sources: G. K. Chesterton, The Thing, 1929 · SEC order on Knight Capital, 2013

next: Murphy's law